Bayesian Optimization Report: Function 2

GP Health Summary per Iteration

Iteration	GP Health	Status
1	0.735	Good
2	0.734	Good
3	0.735	Good
4	0.735	Good
5	0.737	Good
6	0.735	Good
7	0.734	Good
8	0.732	Good
9	0.731	Good
10	0.729	Good
11	0.727	Good
12	0.725	Good
13	0.724	Good
14	0.723	Good
15	0.719	Good
16	0.718	Good
17	0.718	Good
18	0.718	Good
19	0.717	Good
20	0.717	Good
21	0.717	Good
22	0.717	Good
23	0.717	Good
24	0.717	Good
25	0.719	Good
26	0.719	Good
27	0.719	Good
28	0.718	Good
29	0.718	Good
30	0.717	Good

GP Sigma Summary

Iteration	Mean Sigma	Min Sigma	Max Sigma
1	0.1599	0.0637	0.2235
2	0.1490	0.0597	0.2180
3	0.1387	0.0618	0.1908
4	0.1306	0.0625	0.1850
5	0.1243	0.0563	0.1675
6	0.1203	0.0605	0.1629
7	0.1158	0.0630	0.1580
8	0.1138	0.0569	0.1552
9	0.1137	0.0528	0.1536
10	0.1105	0.0578	0.1512
11	0.1084	0.0536	0.1504
12	0.1086	0.0550	0.1492
13	0.1069	0.0542	0.1471
14	0.1047	0.0508	0.1459
15	0.1029	0.0513	0.1425
16	0.1016	0.0424	0.1422
17	0.1012	0.0471	0.1408
18	0.0999	0.0434	0.1388
19	0.0994	0.0436	0.1382
20	0.0963	0.0356	0.1371
21	0.0955	0.0395	0.1351
22	0.0961	0.0378	0.1346
23	0.0943	0.0402	0.1346
24	0.0952	0.0365	0.1329
25	0.1172	0.0325	0.1971
26	0.1134	0.0289	0.1966
27	0.1141	0.0291	0.1949
28	0.1132	0.0287	0.1939
29	0.1167	0.0265	0.1943
30	0.1172	0.0257	0.1957

Best Results Per Iteration

Iteration	Best Input	Best Output	Kernel	Acquisition	GP Health	Anomalies
1	[0.7798454 0.10698064]	0.39018759403612335	Matern(ν=0.5)	UCB	0.735	None
2	[0.24348906 0.99681886]	0.05461568629453417	Matern(ν=0.5)	UCB	0.734	None
3	[0.50616936 0.78118645]	0.154412991881096	Matern(ν=0.5)	UCB	0.735	None
4	[0.07775472 0.04532463]	0.027973884716737396	Matern(ν=0.5)	UCB	0.735	None
5	[0.7503806 0.81485528]	0.42983254010369165	Matern(ν=0.5)	UCB	0.737	None
6	[0.62413543 0.05061774]	0.30287186574524455	Matern(ν=0.5)	UCB	0.735	None
7	[0.68286129 0.66770199]	0.6255466307087834	Matern(ν=0.5)	UCB	0.734	None
8	[0.70244218 0.193888]	0.6689600666050541	Matern(ν=0.5)	UCB	0.732	None
9	[0.68447155 0.01656597]	0.6566710106899065	Matern(ν=0.5)	UCB	0.731	None
10	[0.69417226 0.30508741]	0.6895802617503616	Matern(ν=0.5)	UCB	0.729	None
11	[0.69389842 0.66921803]	0.6590555698403309	Matern(ν=0.5)	UCB	0.727	None
12	[0.71482733 0.98826686]	0.5784485759294972	Matern(ν=0.5)	UCB	0.725	None
13	[0.70799238 0.08634314]	0.6436817145038488	Matern(ν=0.5)	UCB	0.724	None
14	[0.69980013 0.07240104]	0.678338698231876	Matern(ν=0.5)	UCB	0.723	None
15	[0.69587337 0.17615416]	0.6943680625674236	Matern(ν=0.5)	UCB	0.719	None
16	[0.70007375 0.57320913]	0.6665035174461527	Matern(ν=0.5)	UCB	0.718	None
17	[0.6940281 0.02639042]	0.701457514203829	Matern(ν=0.5)	UCB	0.718	None
18	[0.68934185 0.12728525]	0.6958972616031501	Matern(ν=0.5)	UCB	0.718	None
19	[0.69778342 0.39593345]	0.6805723222011436	Matern(ν=0.5)	UCB	0.717	None
20	[0.69108556 0.39561004]	0.681099855532233	Matern(ν=0.5)	UCB	0.717	None
21	[0.69397051 0.16314431]	0.7023069815848528	Matern(ν=0.5)	UCB	0.717	None

22	[0.68685292 0.20800897]	0.6754153219860016	Matern(ν=0.5)	UCB	0.717	None
23	[0.70618116 0.57061294]	0.6710020164321493	Matern(ν=0.5)	UCB	0.717	None
24	[0.69009558 0.03509315]	0.7017333675865385	Matern(ν=0.5)	UCB	0.717	None
25	[0.69410444 0.33459519]	0.6865143896276364	Matern(ν=0.5)	UCB	0.719	None
26	[0.69635749 0.47605379]	0.6910764596130333	Matern(ν=0.5)	UCB	0.719	None
27	[0.69466697 0.22959851]	0.6897568031456767	Matern(ν=0.5)	UCB	0.719	None
28	[0.69178391 0.10491143]	0.7037954481522438	Matern(ν=0.5)	UCB	0.718	None
29	[0.6975539 0.02784764]	0.6963525237397064	Matern(ν=0.5)	UCB	0.718	None
30	[0.69431827 0.01472529]	0.696998467197623	Matern(ν=0.5)	UCB	0.717	None

Acquisition Values Per Iteration

Iteration 1

Method	Max Value
UCB	0.736074
EI	0.010938
PI	0.203475
THOMPSON	0.583590
KG	0.010938
ENTROPY	1.156152
PORTFOLIO	0.801644

Iteration 2

Method	Max Value
UCB	0.586215
EI	0.011336
PI	0.212590
THOMPSON	0.612250
KG	0.011336
ENTROPY	1.063436

PORTFOLIO	0.677087

Iteration 3

Method	Max Value
UCB	0.580806
EI	0.009655
PI	0.194096
THOMPSON	0.582388
KG	0.009655
ENTROPY	1.070729
PORTFOLIO	0.629562

Iteration 4

Method	Max Value
UCB	0.578104
EI	0.008949
PI	0.173824
THOMPSON	0.541745
KG	0.008949
ENTROPY	1.039793
PORTFOLIO	0.578052

Iteration 5

Method	Max Value
UCB	0.579209
EI	0.008874
PI	0.189166
THOMPSON	0.677021
KG	0.008874
ENTROPY	1.068801
PORTFOLIO	0.575102

Iteration 6

Method	Max Value
UCB	0.575736

EI	0.008296
PI	0.178758
THOMPSON	0.572500
KG	0.008296
ENTROPY	1.037188
PORTFOLIO	0.523103

Iteration 7

Method	Max Value
UCB	0.571279
EI	0.007507
PI	0.159585
THOMPSON	0.681958
KG	0.007507
ENTROPY	1.022517
PORTFOLIO	0.514474

Iteration 8

Method	Max Value
UCB	0.571731
EI	0.007749
PI	0.178201
THOMPSON	0.572471
KG	0.007749
ENTROPY	1.022225
PORTFOLIO	0.481184

Iteration 9

Method	Max Value
UCB	0.566151
EI	0.006590
PI	0.170339
THOMPSON	0.623079
KG	0.006590
ENTROPY	1.032482

PORTFOLIO	0.474768

Iteration 10

Method	Max Value
UCB	0.561770
EI	0.005908
PI	0.164291
THOMPSON	0.583082
KG	0.005908
ENTROPY	0.978418
PORTFOLIO	0.443859

Iteration 11

Method	Max Value
UCB	0.556260
EI	0.004733
PI	0.152264
THOMPSON	0.614128
KG	0.004733
ENTROPY	0.965568
PORTFOLIO	0.437081

Iteration 12

Method	Max Value
UCB	0.557965
EI	0.005368
PI	0.166510
THOMPSON	0.547794
KG	0.005368
ENTROPY	0.982248
PORTFOLIO	0.448199

Iteration 13

Method	Max Value
UCB	0.559107

EI	0.005804
PI	0.178763
THOMPSON	0.532738
KG	0.005804
ENTROPY	0.954356
PORTFOLIO	0.409941

Iteration 14

Method	Max Value
UCB	0.543076
EI	0.003092
PI	0.100965
THOMPSON	0.516987
KG	0.003092
ENTROPY	0.944774
PORTFOLIO	0.419398

Iteration 15

Method	Max Value
UCB	0.546357
EI	0.003954
PI	0.133208
THOMPSON	0.577730
KG	0.003954
ENTROPY	0.935504
PORTFOLIO	0.403834

Iteration 16

Method	Max Value
UCB	0.537027
EI	0.002354
PI	0.095066
THOMPSON	0.535651
KG	0.002354
ENTROPY	0.956072

PORTFOLIO	0.388435

Iteration 17

Method	Max Value
UCB	0.548089
EI	0.004297
PI	0.148505
THOMPSON	0.530208
KG	0.004297
ENTROPY	0.944622
PORTFOLIO	0.387556

Iteration 18

Method	Max Value
UCB	0.543376
EI	0.003420
PI	0.125243
THOMPSON	0.582868
KG	0.003420
ENTROPY	0.952819
PORTFOLIO	0.402808

Iteration 19

Method	Max Value
UCB	0.533157
EI	0.001716
PI	0.080915
THOMPSON	0.542795
KG	0.001716
ENTROPY	0.938505
PORTFOLIO	0.375831

Iteration 20

Method	Max Value
UCB	0.532466

EI	0.001937
PI	0.078918
THOMPSON	0.548077
KG	0.001937
ENTROPY	0.929342
PORTFOLIO	0.388143

Iteration 21

Method	Max Value
UCB	0.539678
EI	0.002551
PI	0.116029
THOMPSON	0.551426
KG	0.002551
ENTROPY	0.916929
PORTFOLIO	0.371635

Iteration 22

Method	Max Value
UCB	0.531814
EI	0.001374
PI	0.075613
THOMPSON	0.551303
KG	0.001374
ENTROPY	0.918573
PORTFOLIO	0.383694

Iteration 23

Method	Max Value
UCB	0.522702
EI	0.000981
PI	0.046853
THOMPSON	0.511296
KG	0.000981
ENTROPY	0.909926

PORTFOLIO	0.372402

Iteration 24

Method	Max Value
UCB	0.540996
EI	0.002694
PI	0.131210
THOMPSON	0.531914
KG	0.002694
ENTROPY	0.912024
PORTFOLIO	0.376369

Iteration 25

Method	Max Value
UCB	0.521854
EI	0.000379
PI	0.028820
THOMPSON	0.495613
KG	0.000379
ENTROPY	1.008257
PORTFOLIO	0.352134

Iteration 26

Method	Max Value
UCB	0.526898
EI	0.000725
PI	0.047593
THOMPSON	0.524494
KG	0.000725
ENTROPY	0.980629
PORTFOLIO	0.356976

Iteration 27

Method	Max Value
UCB	0.525003

EI	0.000433
PI	0.035638
THOMPSON	0.519267
KG	0.000433
ENTROPY	1.000494
PORTFOLIO	0.353717

Iteration 28

Method	Max Value
UCB	0.530463
EI	0.000788
PI	0.060085
THOMPSON	0.526815
KG	0.000788
ENTROPY	0.988358
PORTFOLIO	0.355547

Iteration 29

Method	Max Value
UCB	0.525603
EI	0.000489
PI	0.039264
THOMPSON	0.509839
KG	0.000489
ENTROPY	1.001823
PORTFOLIO	0.357707

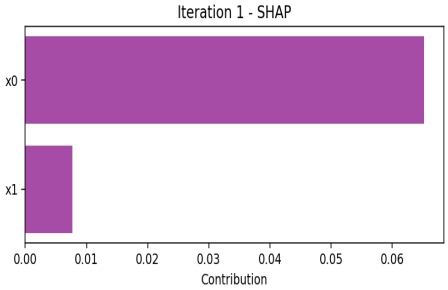
Iteration 30

Method	Max Value
UCB	0.523202
EI	0.000367
PI	0.032518
THOMPSON	0.519018
KG	0.000367
ENTROPY	1.005163

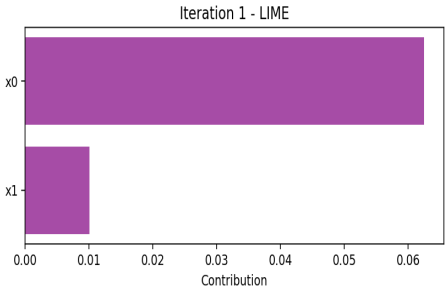
PORTFOLIO	0.358044

Feature Explanations (SHAP / LIME)

Iteration 1 - SHAP



Iteration 1 - LIME



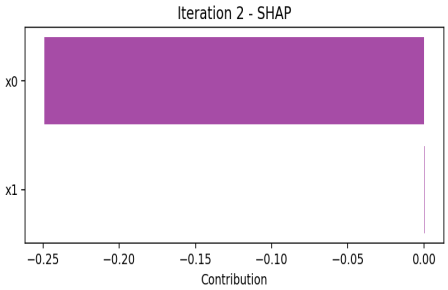
Iteration 1 - SHAP Values

Feature	SHAP Value
x0	0.06519
x1	0.00767

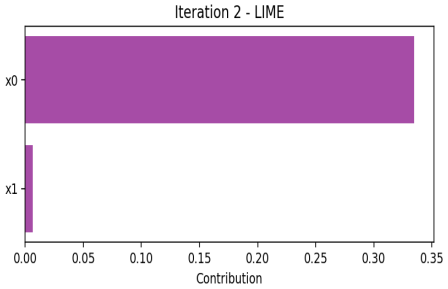
Iteration 1 - LIME Values

Feature	LIME Value
x0	0.06245
x1	0.01002

Iteration 2 - SHAP



Iteration 2 - LIME



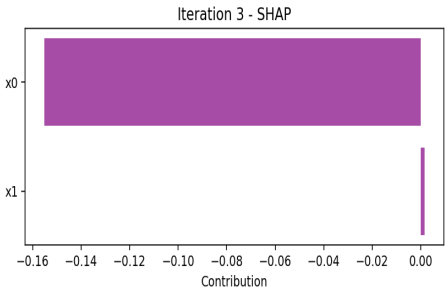
Iteration 2 - SHAP Values

Feature	SHAP Value
x0	-0.24913
x1	0.00052

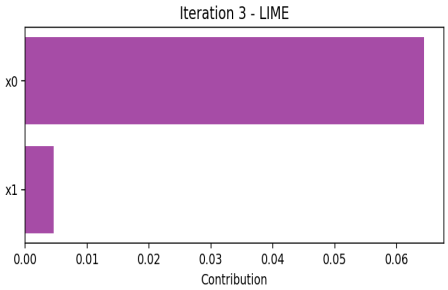
Iteration 2 - LIME Values

Feature	LIME Value
x0	0.33497
x1	0.00630

Iteration 3 - SHAP



Iteration 3 - LIME



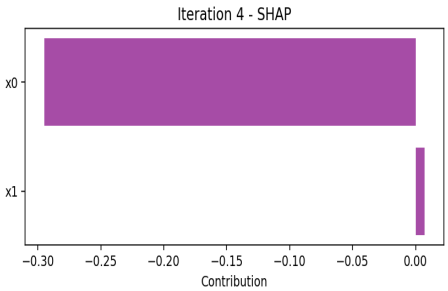
Iteration 3 - SHAP Values

Feature	SHAP Value
x0	-0.15514
x1	0.00159

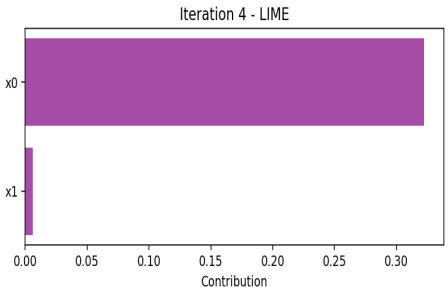
Iteration 3 - LIME Values

Feature	LIME Value
x0	0.06441
x1	0.00459

Iteration 4 - SHAP



Iteration 4 - LIME



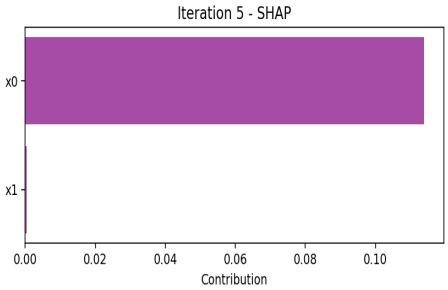
Iteration 4 - SHAP Values

Feature	SHAP Value
x0	-0.29487
x1	0.00730

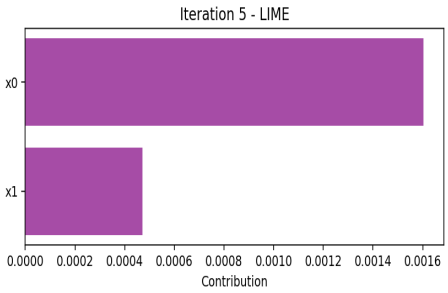
Iteration 4 - LIME Values

Feature	LIME Value
x0	0.32223
x1	0.00618

Iteration 5 - SHAP



Iteration 5 - LIME



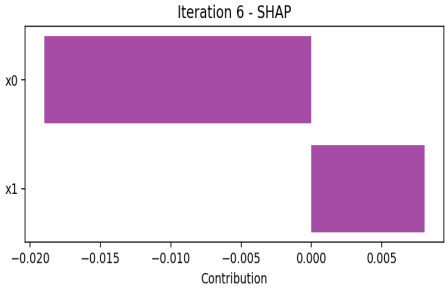
Iteration 5 - SHAP Values

Feature	SHAP Value
x0	0.11384
x1	0.00046

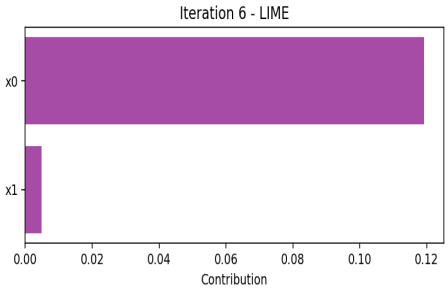
Iteration 5 - LIME Values

Feature	LIME Value
x0	0.00160
x1	0.00047

Iteration 6 - SHAP



Iteration 6 - LIME



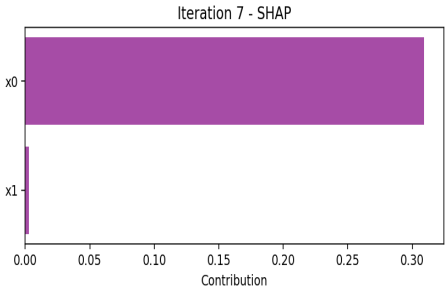
Iteration 6 - SHAP Values

Feature	SHAP Value
x0	-0.01900
x1	0.00806

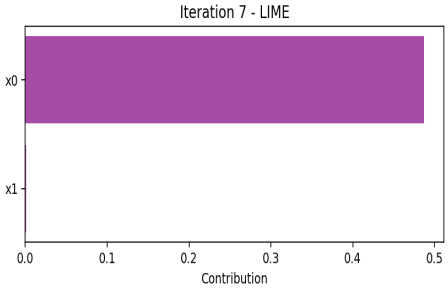
Iteration 6 - LIME Values

Feature	LIME Value
x0	0.11917
x1	0.00485

Iteration 7 - SHAP



Iteration 7 - LIME



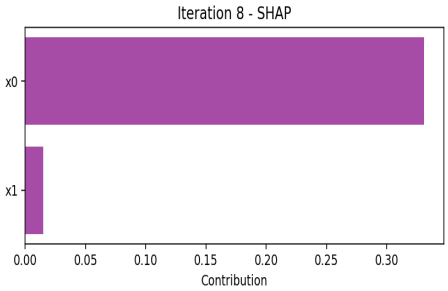
Iteration 7 - SHAP Values

Feature	SHAP Value
x0	0.30900
x1	0.00262

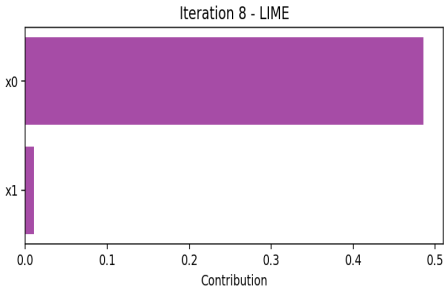
Iteration 7 - LIME Values

Feature	LIME Value
x0	0.48696
x1	0.00131

Iteration 8 - SHAP



Iteration 8 - LIME



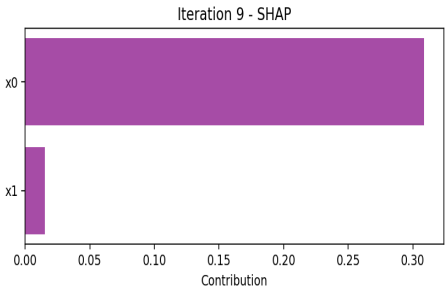
Iteration 8 - SHAP Values

Feature	SHAP Value
x0	0.33089
x1	0.01484

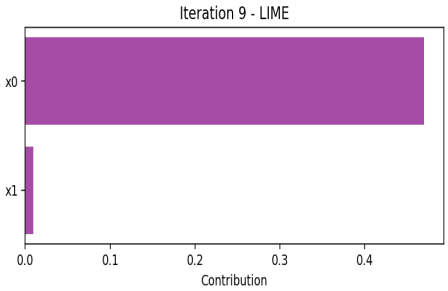
Iteration 8 - LIME Values

Feature	LIME Value
x0	0.48572
x1	0.01052

Iteration 9 - SHAP



Iteration 9 - LIME



Iteration 9 - SHAP Values

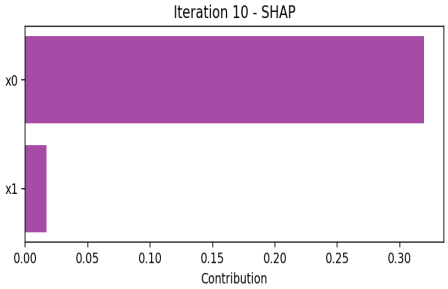
Feature	SHAP Value
x0	0.30860
x1	0.01513

Iteration 9 - LIME Values

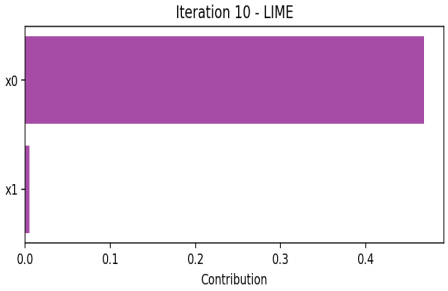
Feature	LIME Value
x0	0.46953

x1	0.00921

Iteration 10 - SHAP



Iteration 10 - LIME



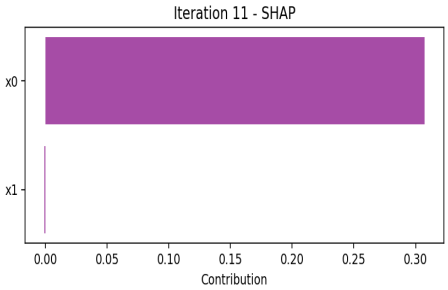
Iteration 10 - SHAP Values

Feature	SHAP Value
x0	0.31943
x1	0.01694

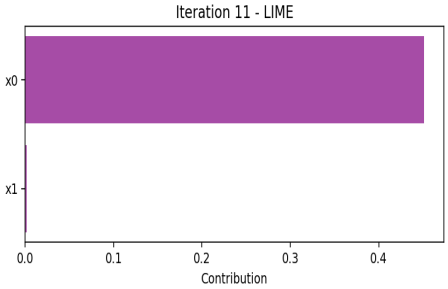
Iteration 10 - LIME Values

Feature	LIME Value
x0	0.46824
x1	0.00474

Iteration 11 - SHAP



Iteration 11 - LIME



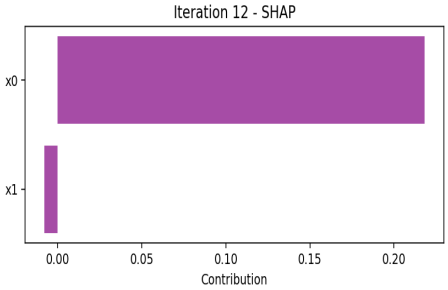
Iteration 11 - SHAP Values

Feature	SHAP Value
x0	0.30751
x1	-0.00090

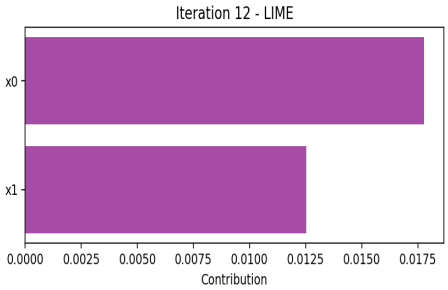
Iteration 11 - LIME Values

Feature	LIME Value
x0	0.45121
x1	0.00151

Iteration 12 - SHAP



Iteration 12 - LIME



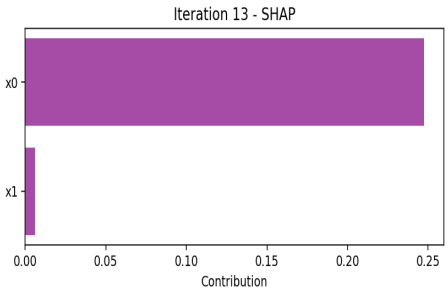
Iteration 12 - SHAP Values

Feature	SHAP Value
x0	0.21838
x1	-0.00801

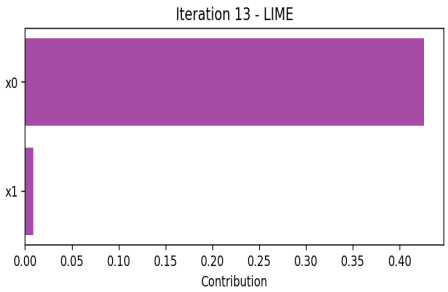
Iteration 12 - LIME Values

Feature	LIME Value
x0	0.01777
x1	0.01254

Iteration 13 - SHAP



Iteration 13 - LIME



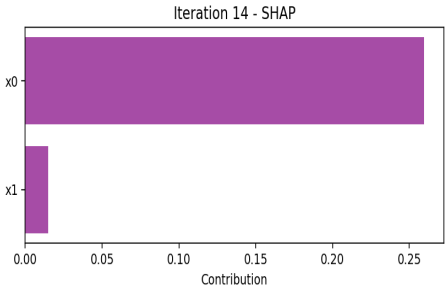
Iteration 13 - SHAP Values

Feature	SHAP Value
x0	0.24743
x1	0.00594

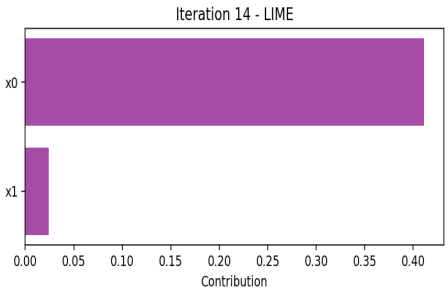
Iteration 13 - LIME Values

Feature	LIME Value
x0	0.42579
x1	0.00850

Iteration 14 - SHAP



Iteration 14 - LIME



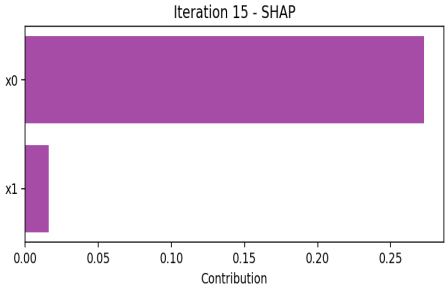
Iteration 14 - SHAP Values

Feature	SHAP Value
x0	0.25944
x1	0.01470

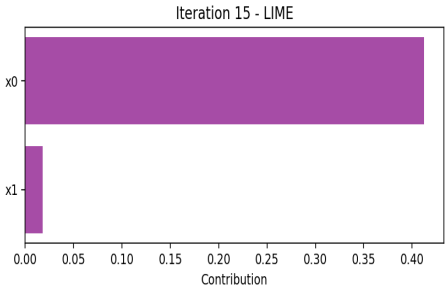
Iteration 14 - LIME Values

Feature	LIME Value
x0	0.41100
x1	0.02388

Iteration 15 - SHAP



Iteration 15 - LIME



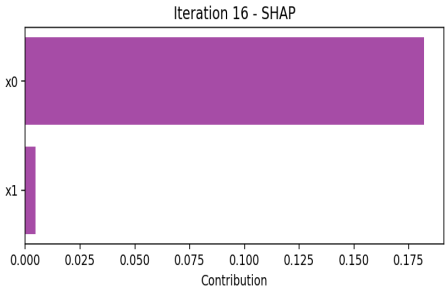
Iteration 15 - SHAP Values

Feature	SHAP Value
x0	0.27268
x1	0.01610

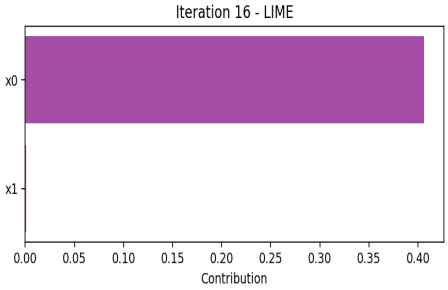
Iteration 15 - LIME Values

Feature	LIME Value
x0	0.41221
x1	0.01803

Iteration 16 - SHAP



Iteration 16 - LIME



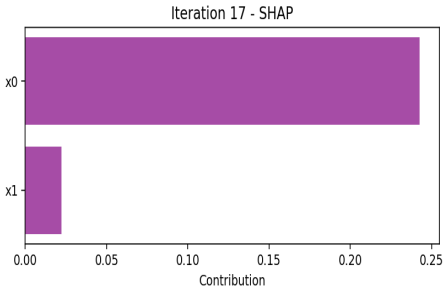
Iteration 16 - SHAP Values

Feature	SHAP Value
x0	0.18193
x1	0.00458

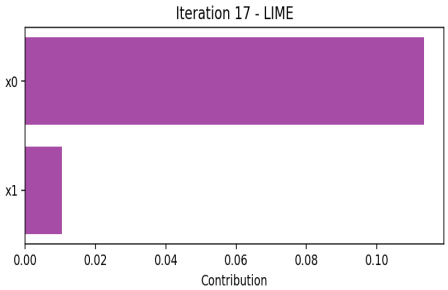
Iteration 16 - LIME Values

Feature	LIME Value
x0	0.40598
x1	0.00088

Iteration 17 - SHAP



Iteration 17 - LIME



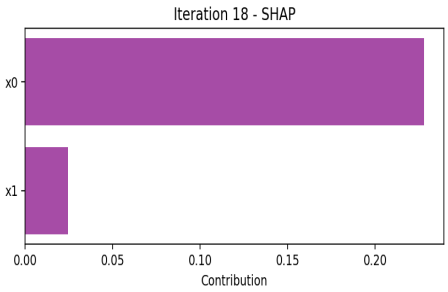
Iteration 17 - SHAP Values

Feature	SHAP Value
x0	0.24260
x1	0.02235

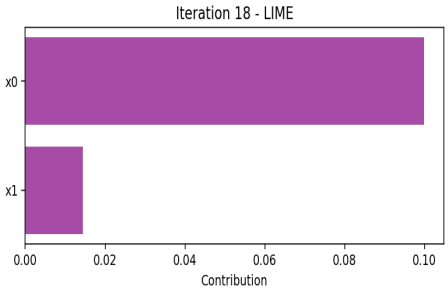
Iteration 17 - LIME Values

Feature	LIME Value
x0	0.11342
x1	0.01038

Iteration 18 - SHAP



Iteration 18 - LIME



Iteration 18 - SHAP Values

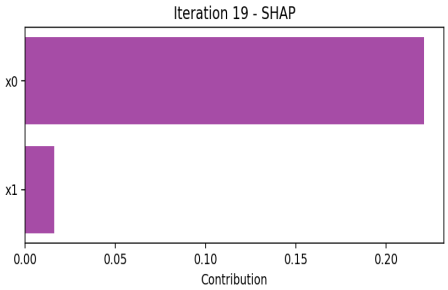
Feature	SHAP Value
x0	0.22793
x1	0.02438

Iteration 18 - LIME Values

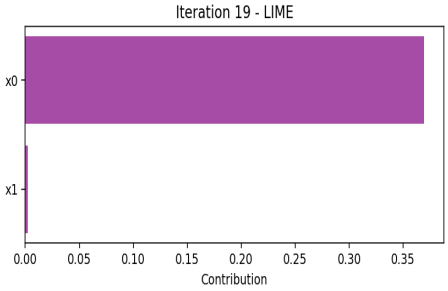
Feature	LIME Value
x0	0.09986

x1	0.01440

Iteration 19 - SHAP



Iteration 19 - LIME



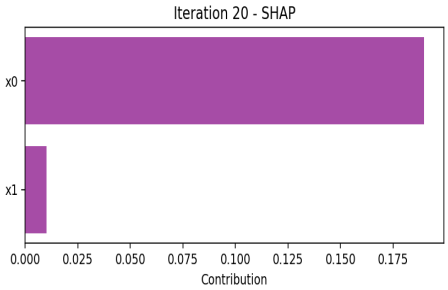
Iteration 19 - SHAP Values

Feature	SHAP Value
x0	0.22087
x1	0.01609

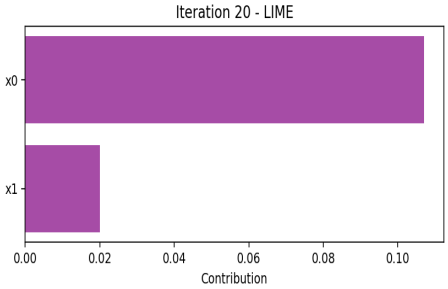
Iteration 19 - LIME Values

Feature	LIME Value
x0	0.36926
x1	0.00227

Iteration 20 - SHAP



Iteration 20 - LIME



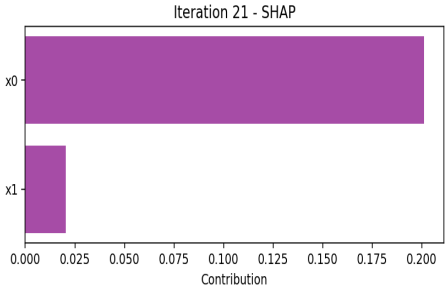
Iteration 20 - SHAP Values

Feature	SHAP Value
x0	0.18970
x1	0.01000

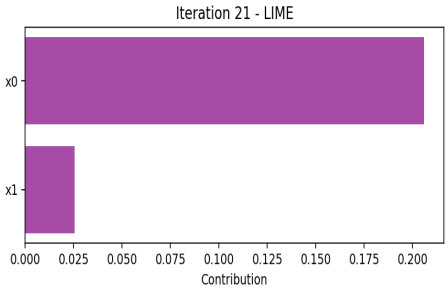
Iteration 20 - LIME Values

Feature	LIME Value
x0	0.10699
x1	0.02006

Iteration 21 - SHAP



Iteration 21 - LIME



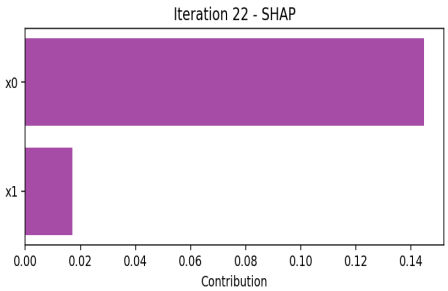
Iteration 21 - SHAP Values

Feature	SHAP Value
x0	0.20112
x1	0.02054

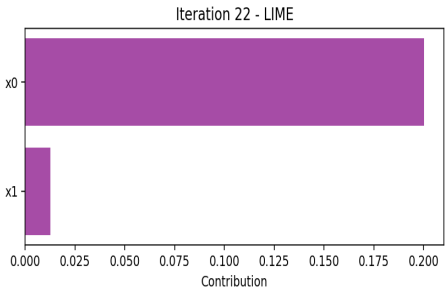
Iteration 21 - LIME Values

Feature	LIME Value
x0	0.20583
x1	0.02552

Iteration 22 - SHAP



Iteration 22 - LIME



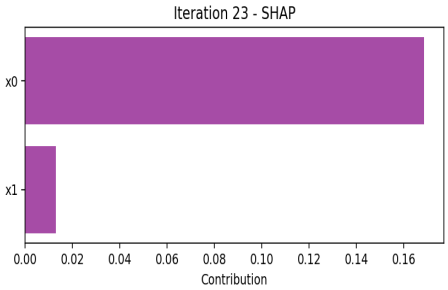
Iteration 22 - SHAP Values

Feature	SHAP Value
x0	0.14470
x1	0.01706

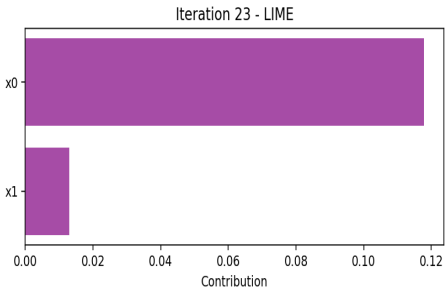
Iteration 22 - LIME Values

Feature	LIME Value
x0	0.20011
x1	0.01271

Iteration 23 - SHAP



Iteration 23 - LIME



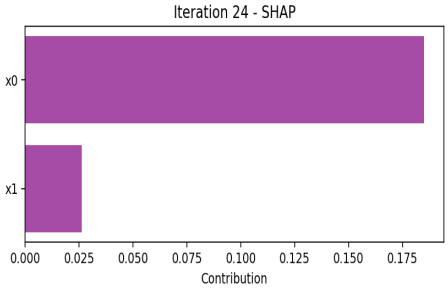
Iteration 23 - SHAP Values

Feature	SHAP Value
x0	0.16844
x1	0.01290

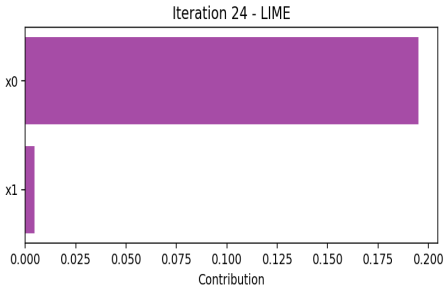
Iteration 23 - LIME Values

Feature	LIME Value
x0	0.11781
x1	0.01293

Iteration 24 - SHAP



Iteration 24 - LIME



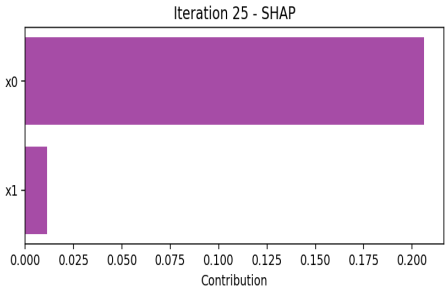
Iteration 24 - SHAP Values

Feature	SHAP Value
x0	0.18490
x1	0.02607

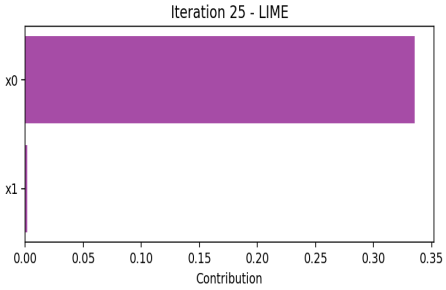
Iteration 24 - LIME Values

Feature	LIME Value
x0	0.19494
x1	0.00459

Iteration 25 - SHAP



Iteration 25 - LIME



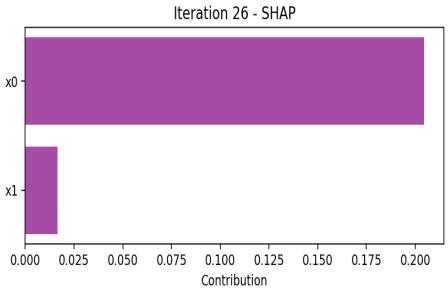
Iteration 25 - SHAP Values

Feature	SHAP Value
x0	0.20609
x1	0.01121

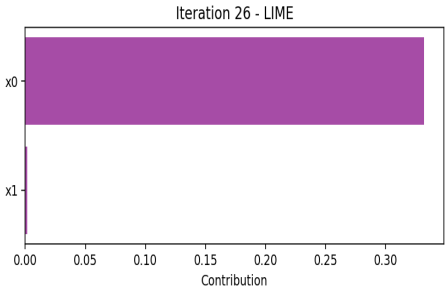
Iteration 25 - LIME Values

Feature	LIME Value
x0	0.33531
x1	0.00182

Iteration 26 - SHAP



Iteration 26 - LIME



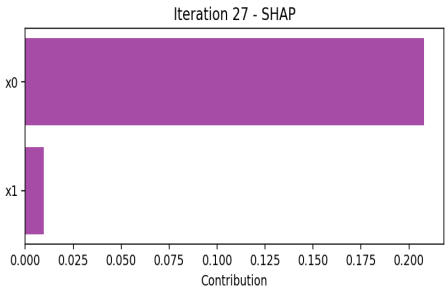
Iteration 26 - SHAP Values

Feature	SHAP Value
x0	0.20443
x1	0.01640

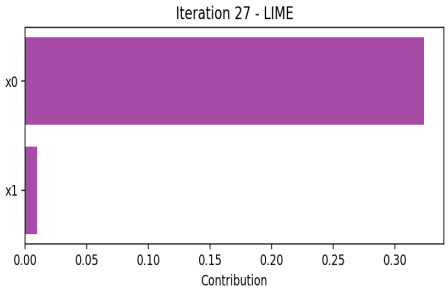
Iteration 26 - LIME Values

Feature	LIME Value
x0	0.33172
x1	0.00142

Iteration 27 - SHAP



Iteration 27 - LIME



Iteration 27 - SHAP Values

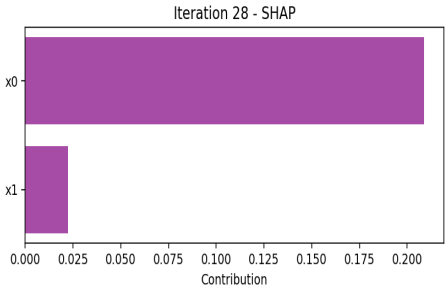
Feature	SHAP Value
x0	0.20780
x1	0.00964

Iteration 27 - LIME Values

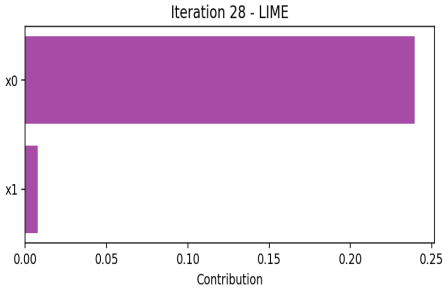
Feature	LIME Value
x0	0.32334

x1	0.00969

Iteration 28 - SHAP



Iteration 28 - LIME



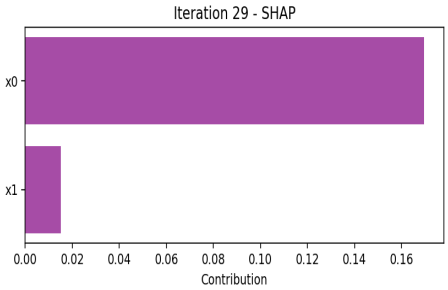
Iteration 28 - SHAP Values

Feature	SHAP Value
x0	0.20877
x1	0.02225

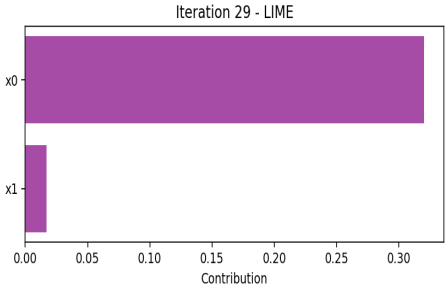
Iteration 28 - LIME Values

Feature	LIME Value
x0	0.23964
x1	0.00768

Iteration 29 - SHAP



Iteration 29 - LIME



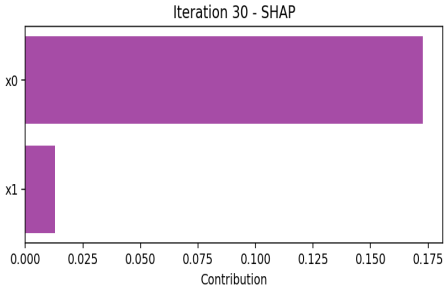
Iteration 29 - SHAP Values

Feature	SHAP Value
x0	0.16948
x1	0.01515

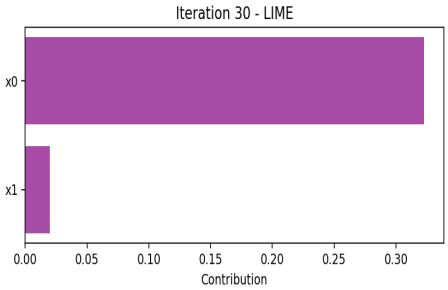
Iteration 29 - LIME Values

Feature	LIME Value
x0	0.32009
x1	0.01707

Iteration 30 - SHAP



Iteration 30 - LIME



Iteration 30 - SHAP Values

Feature	SHAP Value
x0	0.17274
x1	0.01289

Iteration 30 - LIME Values

Feature	LIME Value
x0	0.32260
x1	0.01988